From a Single-Tank to a Scale-Incorporation Model

Where the six-parameter replacement model strains, and a portfolio sketch that fixes it

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Purpose

This note is a design sketch, not a results report. It does two things:

1. Sets out the **worries** with the current six-parameter (6p) replacement model — the ways its single-tank state abstraction strains against the multi-tank portfolio dynamics in the data, with the evidence from the Ticket 013 (beginning-of-year, BOY) diagnostics.
2. Sketches a **scale-incorporation model**: a facility-level dynamic discrete choice model whose state is the facility's *portfolio of tanks* rather than a single representative tank, with a size-specific operating payoff and an explicit partial-closure / downsizing margin.

Nothing here changes the estimator or the live pipeline. It is the architect-level economic design to be turned into an implementation ticket only after the open empirical checks in Section 6 are run.

A unit reminder carried throughout: one model unit = \$10,000 per year. The current model normalizes the operating payoff to one such unit per *facility*; the scale model re-bases this to a payoff per unit of *tank capacity* (Section 4.3), which is the load-bearing change for welfare.

The six-parameter model in one paragraph

The 6p model treats a facility as a **single tank** moving through a discretized state: an age bin $b \in \{1, \dots, 8\}$, a wall type $w \in \{\mathrm{SW}, \mathrm{DW}\}$, and a regime $g \in \{\mathrm{FF}, \mathrm{RB}\}$ ($8 \times 2 \times 2 = 32$ cells). Each year the facility chooses one of three actions: **maintain**, **exit**, or **replace** (install a new tank, which resets age to bin 1). Flow utility while operating is

$$u_M \;=\; \underbrace{1}_{R,\ \text{normalized}} \;+\gamma_p\,P \;-\; \gamma_r\,h, \qquad u_E = \kappa, \qquad u_R = -K,$$

with the risk-based premium P and expected hazard loss h entering the maintain payoff only. This is the textbook Rust architecture: a scalar state discretized into bins, with one unit (here, one facility) occupying one bin.

Where it strains: worries with the single-tank abstraction

The problem is that a facility is **not** one tank. It is a portfolio of tanks of different ages, wall types, and sizes. The 6p model collapses that portfolio to a single **average-age** cell, and that collapse is the source of every worry below.

The average-age collapse is lossy and non-monotonic

A facility with one brand-new tank and one thirty-year-old tank is placed at a *middle* age bin — it looks like a homogeneous, middle-aged single tank. The old tank, which is the hazard and removal candidate, is invisible; the hazard is computed off the wrong (averaged) age. Worse, as tanks are added or pruned the average moves **non-monotonically** across years even when no individual tank's status changed, generating spurious state transitions.

Replace transitions are almost never on-support

The single-tank `F_replace` resets the facility's *average* age to bin 1. But a one-tank swap at a multi-tank facility barely moves the average, so the modeled post-replacement state rarely matches the data. The Ticket 013 diagnostics quantify this: **98.3% of replacement transitions land off-support** (5,310 of 5,404). The replacement margin — the very object the model exists to study — is essentially never represented correctly for multi-tank facilities.

Partial closure is mislabeled as maintain

Closing *some* tanks while continuing to operate ("partial closure" or downsizing) has no action in the 6p model. It gets folded into **maintain**, and shows up as an average-age jump under a maintain label. After BOY stamping fixed the wall dimension, the residual maintain off-support is **0.685%** (14,652 rows), of which 14,121 are exactly these partial-closure age jumps. Small in mass, but it is a *structural* gap, not a coding error: the single-tank maintain transition cannot represent it.

The double-walled replace margin: degeneracy resolved, churn-vs-replacement remains

Under the earlier end-of-year stamping the DW conditional-replace share was pinned at one (Pr(replace | close) = 1 for all DW cells), because DW exits — a DW facility has zero tanks at year-end — were routed out of the DW cells by the worst-tank rule. **Beginning-of-year stamping (Ticket 013) resolved that degeneracy:** the DW cells now retain their exits (all-DW closures split 3,773 exit / 745 replace, so Pr(replace | close) ≈ 0.17), and K_{DW} is now identified off real within-DW variation rather than off the close rate and functional form alone. What the single-tank model *still* cannot do is separate a genuine like-for-like replacement from within-facility tank churn (one DW tank swapped at a large, DW-heavy facility): that separation is the replacement transition itself, which remains 98% off-support (above), and only the portfolio state recovers it.

It cannot generate the size-dependent regime response

The headline reduced-form fact is that the flat-fee-to-risk-based (FF→RB) response is **size-dependent and sign-reversing**: about +6.2 percentage points at facilities with 4+ tanks, but −4.9 to −6.9 points at 2–3 tank facilities. A single-tank model facing a common regime shock can scale a response by size but cannot **flip its sign**. A sign reversal points to a margin that exists at large facilities and not at small ones — partial downsizing — which the 6p model does not contain.

Consequence for the counterfactuals

The four target counterfactuals are not equally exposed. The replacement-subsidy CF leans directly on the margin that is 98% off-support, and the size-HTE ("Texas stays flat-fee") and age-mandate CFs need per-tank age that the average collapses. The table summarizes the exposure.

Table 1: Counterfactual exposure to the single-tank abstraction.

Counterfactual	Margin it leans on	6p exposure
TX stays flat-fee (size HTE)	size-dependent close/replace	high (sign reversal)
Pigouvian / first best	per-tank marginal damage	low (additive premium)
Replacement subsidy curve	replacement transition	**high** (98% off-support)
Removal mandate at age A	per-tank age at threshold	high (average collapse)

Notes: Pigouvian is least exposed because the risk-based premium is computed per tank and summed to the facility, so its internalization is additive and framing-invariant. The other three each require a portfolio or per-tank object the single-tank state does not carry.

The portfolio model

The model exists to answer four counterfactuals: Texas stays flat-fee; a first-best Pigouvian contract; a single-walled replacement subsidy; and an age-A removal mandate. It is judged by coverage of the margins those counterfactuals move, not by goodness of fit. Every design choice below traces to an exhibit in Part 3 or a counterfactual requirement in Part 4.

State

$$s = (n,\, G,\, \rho), \qquad n \in \mathbb{N}_0^{16}$$

n is a vector of tank counts over 16 categories formed by wall type (single- or double-walled) crossed with eight five-year age bins; G is the facility's capacity quartile bin; ρ is the insurance regime, fixed over the facility's life. The owner thinks about tanks the way the insurer prices them and leaks happen — through construction type and age — and thinks about scale in gallons. The model's bookkeeping records this as counts per category: tanks within one (wall, age-bin) category are interchangeable (evidence: X5).

First-stage objects

All of the following are measured from data; none are estimated in the NPL step.

- **Empirical cell prices $\bar{p}_c$.** The average premium actually paid by single-cell Texas facility-years (all tanks in one (wall, age-bin) category), 2006 era; one price per cell (X3 empirical card table).
- **FF contract lookup.** Each control state's own flat-fee contract (τ_s, D_s) — per-tank fee and deductible — time-averaged for stationarity. Texas has no registry-recorded per-tank fee; D_{TX} = \$5,000 is assumed. **Kansas and Maryland are excluded from estimation**, on the evidence of the ASTSWMO state-fund surveys the premium file itself cites: both states' funds functioned as *cleanup* funds but did not satisfy federal financial responsibility — Kansas operators were required to carry private third-party liability coverage (K.S.A. 65-34,117), and the ASTSWMO 2002 Table 2 footnote states Maryland owners "need private insurance or be self-insured." The operator-facing premium in both states is therefore private and unrecorded. Kansas is a noted robustness candidate for re-inclusion: its fund deductible *is* recorded (\$3,000 plus \$500 per tank — a deductible with a count slope), usable under the stated assumption that its private liability premium is composition-invariant and absorbed by the Kansas fixed effect.[1]

[1]Per-state contracts cost no parameters and no extra computation: the engine already solves per state for the fixed effects, and the contract variation survives the state fixed effects because contracts enter multiplied by within-state-varying quantities — τ_s as a slope on $N(n)$, D_s as a slope on $H(n)$. An additive fixed effect absorbs levels, not slopes.

Identification threats. The slope-based identification compares hazard- and count-responsiveness across states with different contracts, so it is exposed to anything that makes those responsiveness differences for other reasons. Three concrete threats: (i) contract terms may be set in response to the state's tank stock — a state with riskier old tanks choosing a higher deductible bends the slope comparison (reverse causality); (ii) enforcement and inspection intensity may co-vary with contract stringency, so behavior attributed to the deductible could partly reflect inspections; (iii) fund payout reliability differs across states — a slow- or unreliably-paying fund changes the owner's effective exposure beyond the stated D. Mitigations, stated without overselling: state fixed effects absorb level confounders, year fixed effects absorb common shocks, and the regime channel is cross-validated by the DiD; threats (i)–(iii) are why the welfare results lean hardest on the within-Texas and regime-contrast variation, with the cross-state slopes as supporting evidence. Zero-fee states still contribute to $\hat{\gamma}_r$ through their deductibles.

- **Facility-level hazard h^{ML}.** The leak data is facility-level and cannot support per-tank rates; the assumption is stated as such.
- **Per-tank aging kernel Λ.** Empirical age-bin transition probabilities (X4 age-transition table).
- **Capacity kernel $\Gamma_G(\cdot \mid G,\, m-k)$.** Empirical capacity-bin transition probabilities, conditioned on net tank change $m-k$ (X4 G-transition table; to be re-estimated on this conditioning in the estimator first stage — removing 3 and installing 2 is not the same event as removing 4 and installing 1).
- **Removal rule $r(n,k)$.** Defined in Section 4.4 below.

Perceived quantities

$$N(n) = \sum_c n_c \qquad P(n,\rho) = \begin{cases} \sum_c n_c\, \bar{p}_c & \rho = \mathrm{RB} \\ \tau_s\, N(n) & \rho = \mathrm{FF} \end{cases}$$

$$H(n) = h^{ML}(\bar{a}(n),\, w(n)) \qquad OOP(n) = H(n) \cdot \min\{D,\, L\}$$

P is the actual average insurance bill by construction — empirical prices under risk-based pricing, a per-tank flat fee under flat-fee. H is "a leak at my facility this year," read off the risk model at the portfolio's average-age band and majority wall type — no finer than the data support. OOP is the contract: when a leak occurs, the owner pays the deductible up to the cleanup cost; the minimum is nearly always D since cleanup $L \approx \$300{,}000$.

Actions: the (k, m) menu

$$\mathcal{A}(s) = \{(k, m) : 0 \le k \le N(n),\ 0 \le m \le \bar{m}\} \cup \{X\}$$

$(0, 0)$ is maintain; $(k, 0)$ with $k > 0$ is downsize; $(k, m \ge 1)$ is replace or consolidate; X is full exit. Expansion is outside the model (fenced; evidence: X1).

The menu works like wings priced per wing: each (k, m) is just another option, and two per-unit costs price every line; the model never predicts k or m, the owner chooses them. Only 13.7% of replacements remove exactly one tank; the modal replacement removes 3 and installs 2 (X7b). The removal rule identifies which tanks leave:

the k removed tanks are the oldest single-walled; if fewer than k exist, the oldest double-walled fill the remainc

$$n' = n - r(n, k) + m\, e_{DW,1}$$

New installs are double-walled by law — a regulation enforced in every sample state and confirmed in X9.

Flow payoffs

Sign convention. All money outflows carry explicit minus signs. γ_p and γ_r are positive dollar-to-utility exchange rates. The restriction $\gamma_p = \gamma_r$ — that a certain premium dollar and an expected out-of-pocket dollar enter utility identically — is a testable restriction the data can reject; differences indicate salience effects or risk-weighting.

For an operating action with post-action composition n':

$$u_{(k,m)}(s) = \varphi_G + \alpha_g - \gamma_p\, P(n', \rho) - \gamma_r\, OOP(n') - k\, c^{rem} - m\, c^{inst}$$

For full exit:

$$u_X(s) = \kappa_1\, N(n), \qquad \kappa_0 := 0$$

Setting $\kappa_0 := 0$ is a units choice: it pins the scale of the value function and frees φ_G to measure the level of operating payoffs in model units.

Laws of motion

The k tanks identified by $r(n,k)$ leave; m new double-walled tanks enter at age bin 1. Each surviving tank advances one bin at year-end via the aging kernel Λ; the bin-8 absorbing state captures tanks that age out of the grid. Capacity G transitions via $\Gamma_G(\cdot \mid G,\, m-k)$; it is degenerate at $(0,0)$. Full exit is absorbing; ρ is fixed. Every object here is an exhibit in Part 3, not an assumption.

Choice structure

$$v_a(s) = u_a(s) + \beta \sum_{s'} F_a(s' \mid s)\, V(s'), \qquad \beta = 0.9957$$

$$IV(s) = \lambda \log \sum_{(k,m)\neq(0,0)} e^{v_{(k,m)}(s)/\lambda}$$

$$V(s) = \log\Big(e^{v_{(0,0)}(s)} + e^{IV(s)} + e^{v_X(s)}\Big)$$

$$P((k,m) \mid s) = \frac{e^{IV(s)}}{e^{v_{(0,0)}} + e^{IV} + e^{v_X}} \times \frac{e^{v_{(k,m)}/\lambda}}{\sum_{(k',m')\neq(0,0)} e^{v_{(k',m')}/\lambda}}$$

Each option carries an idiosyncratic draw — what the owner knows that we don't (a retirement plan, a lease renewal, a contractor quote); the owner is rational, the randomness is our ignorance. The three top-level options (maintain, any tank-work, exit) each have an independent draw; the tank-work options share an additional draw parameterized by λ, reflecting the common circumstances that make an owner willing to touch their tanks at all (the same contractor visit, the same permit window). $\lambda \in (0,1]$; at $\lambda = 1$ the model collapses to a flat logit over all options — a special case the data can choose. The menu's payoff structure already makes nearby (k,m) pairs close substitutes; λ carries only the residual shared-circumstances channel beyond what payoffs imply.

Parameters and estimation

$$\theta = \big(\varphi_G,\ \gamma_p,\ \gamma_r,\ c^{rem},\ c^{inst},\ \kappa_1,\ \lambda,\ \{\alpha_g\}\big)$$

$$\hat{\theta} = \arg\max_{\theta} \sum_{i,t} \log P(a_{it} \mid s_{it};\, \theta)$$

Estimated by nested pseudo-likelihood (NPL); the inclusive-value machinery is already in the library. Identification, one line each:

- γ_p: contract contrasts — regime gap \$709 at matched composition (X11); re-filings; cross-state per-tank fees.
- γ_r: hazard variation times deductible levels — within Texas D is flat so identification runs through $H(n)$ varying with composition; cross-state D moves OOP levels.
- $\gamma_p = \gamma_r$: testable.
- c^{rem}, c^{inst}: the (k, m) frequency distribution (X7b).
- κ_1: exit-size gradient — large facilities exit less (odds ratio 0.48; X2).
- λ: co-movement of tank-work options in excess of what observed payoffs explain.

Eight economics parameters plus state fixed effects $\{\alpha_g\}$ on a state space two orders of magnitude richer than the 6p model — parameters did not grow because the removal identity is predictable and costs are per-unit.

Untestable and conventional assumptions

- Shock distribution shape (type-1 extreme value at every nest level).
- $\beta = 0.9957$ — the value carried by the canonical fit's configuration, which is authoritative over any stale documentation value (conventional; sensitivity in Part 4).
- Owners' beliefs equal the estimated hazard and aging processes.
- Stationarity: no anticipation of ban dates or policy changes.
- Facility-level hazard resolution: the leak data cannot support per-tank rates (data limitation, stated explicitly).
- Linear utility in money.
- $\kappa_0 := 0$: units normalization.

The welfare wedge

A leak at facility i costs society $L+E$: cleanup cost L plus external damages E beyond cleanup. Under contract (P, D), the owner bears $\min\{D, L\} = D$ directly; the insurer or fund bears $L - D$. Under risk-based pricing, $L - D$ is priced back into P to the extent the card tracks the facility's own hazard. Under a flat fee, $L - D$ is socialized and does not vary with the facility's hazard. The external damage E is borne by nobody in the market under either regime.

The owner's private marginal cost of own-hazard $H(n)$:

Risk-based. Differentiating $\gamma_p P + \gamma_r OOP$ with respect to H: the facility internalizes D directly through the OOP channel, plus $L - D$ through the actuarial premium channel to the extent the card tracks H. The X3 flatness finding — the empirical schedule is much compressed relative to the filed card — implies this pass-through is partial.

Flat-fee. The per-tank fee τ_s does not vary with H; the facility internalizes only $\gamma_r \cdot D \cdot \partial H / \partial\text{action}$. The term $L - D$ and all of E are fully external.

The first-best counterfactual (CF2) sets the facility's total private exposure equal to $H(n)(L+E)$ at the margin. The wedge policy can close:

- FF wedge per unit hazard: $(L - D) + E$.
- RB wedge: E plus the under-pass-through of $(L - D)$ implied by the compressed empirical card.

Welfare tables report SocialWelfare = ProducerSurplus − ExternalDamage − GovtOutlay under each contract, with E calibrated at $E = \$50{,}000$ per leak event and a band reported around the calibration.

Bringing in revenue: the firm's economic environment as a state

This section is a design addition, not yet in the estimator. It sets out how the firm's revenue — the money it earns by operating — enters the model, what that buys, and how two dynamic-demand papers (Gowrisankaran and Rysman 2012; Sweeting 2013) did the same thing, so the construction here is not invented from scratch.

The gap, and the goal

In the model above the operating payoff is a fixed level, $\varphi_G + \alpha_g$ — it does not move with the economy. That limits what we can learn. The model turns utility into dollars only through the two insurance terms: the premium (γ_p) and the deductible-driven out-of-pocket cost (γ_r). Every dollar weight, and every welfare number, therefore rests on how firms react to insurance prices alone.

The goal is to measure how much a firm's behavior responds to each dollar it faces — a dollar of premium, a dollar of expected out-of-pocket cost, and a dollar of revenue — and to let the data say how big each response is. We are not trying to test whether the firm values a dollar of each the same: that equality is something we would report if we found it, and "force them all equal" is a counterfactual, not the estimation target. But to measure a revenue response at all, the model needs a measure of revenue that varies independently of insurance. That is what this section adds.

What enters, and what it does

Let $\tilde{R}$ be the firm's net operating margin — the money earned per period from operating, built from what we can observe: a state-level fuel margin (retail − wholesale − tax) scaled by volume, or a county-level dollar proxy for local gas-retail activity. The firm is a price-taker on fuel margins, so $\tilde{R}$ is exogenous to its own tank decisions.

$\tilde{R}$ joins the state and the operating payoff:

$$s = (n,\, G,\, \rho,\, \tilde{R}), \qquad u_{(k,m)}(s) = \varphi_G + \alpha_g + \psi\, \tilde{R} - \gamma_p P(n', \rho) - \gamma_r\, OOP(n') - k\, c^{rem} - m\, c^{inst}.$$

It evolves on its own, not in response to what the firm does, so its kernel is action-independent and factors out of the transition:

$$\tilde{R}' \sim \Gamma_R(\cdot \mid \tilde{R}), \qquad F_a(s' \mid s) = F_a(n', G' \mid n, G)\, \Gamma_R(\tilde{R}' \mid \tilde{R}).$$

The choice-specific value then carries one extra sum over $\tilde{R}'$ and leaves the existing (n, G) machinery untouched:

$$v_a(s) = u_a(s) + \beta \sum_{\tilde{R}'} \Gamma_R(\tilde{R}' \mid \tilde{R}) \sum_{n', G'} F_a(n', G' \mid n, G)\, V(s').$$

This is the same kind of object as age and capacity: a state variable with a transition that shifts the value of operating. It is in fact simpler than capacity, whose kernel $\Gamma_G(\cdot \mid G, m-k)$ depends on the action, while revenue's does not. Because $\tilde{R}$ is persistent and the firm forecasts it through Γ_R, it moves the **continuation** value: the value of staying open versus exiting now depends on the expected path of revenue. That is the channel we want — when local economics are good and expected to stay good, marginal facilities stay; when they sour, they exit or shrink.

What this buys:

- A revenue weight ψ that the data — not a normalization — pins down, alongside the insurance weights γ_p and γ_r. We report how big each is.
- An independent source of dollar variation for the cost parameters, so they are no longer identified off insurance alone (the original critique).
- Welfare in real dollars rather than relative to the normalized operating unit.

Calendar time never enters the value function: the model stays stationary because $\tilde{R}$ is a level with a transition, and the year is used only to look up which $\tilde{R}$ a facility-year sits in.

How others brought outside data in: three jobs

Two dynamic-demand papers show that outside data can play three distinct roles in a dynamic model. The construction here uses all three.

Job 1 — a state the agent reacts to and plans around. Gowrisankaran and Rysman (2012) put the things consumers respond to — prices and quality — inside the dynamic state, and have consumers forecast where those are headed; a consumer waits to buy because she expects prices to fall. That forecasting is what makes observed data move continuation values. Their tractability problem is that the full environment is high-dimensional, so they show the consumer needs to track only a single summary number (the "inclusive value") and forecast that one number, not the whole environment. Our $\tilde{R}$ with its transition Γ_R is the same idea in a smaller setting; if adding the revenue state ever makes the state space too large, their inclusive-value reduction (or a value-function approximation) is the known fix.

Job 2 — a payoff piece measured outside the model and plugged in. Sweeting (2013) does not estimate station revenue inside his dynamic model. He measures it first, from outside data — who listens, by demographic, times what advertisers pay per listener — to get revenue as a function of the state, and inserts that into the dynamic step, which then estimates only the cost parameters. This is exactly the architecture this model already uses for premiums, hazard, and the transition kernels: measure the first-stage objects, then let NPL estimate the rest. So building $\tilde{R}$ as a first-stage object — revenue-as-a-function-of-state, computed from the fuel-margin and county data — fits the existing machine rather than changing it.

Job 3 — an extra moment to discipline a weak parameter. Both papers add a moment when the main data under-identifies something. Gowrisankaran and Rysman add an aggregate adoption-rate ("penetration") moment, because total sales alone cannot separate a saturated market from a high-price one; the extra moment is what pins their price-sensitivity heterogeneity. Sweeting matches switching-rate moments — for example, how fast stations flip to a Spanish-language format in heavily-Hispanic versus other markets. For us the same tool applies to the revenue weight ψ: the panel is short (Texas 2006–2020, controls 1999–2020), so if the year-to-year movement in $\tilde{R}$ is too thin to pin ψ on its own, we add a cross-sectional moment from the county data — for instance, that exit is lower where local gas-retail dollars are higher. The state (Job 1, built as a first-stage object, Job 2) and the disciplining moment (Job 3) are complements, not alternatives.

Identification of the revenue weight, and the trend problem

ψ is identified from two sources: the within-state movement of the margin over the panel (time series), and differences in local gas-retail dollars across facilities at a point in time (cross-section — the thicker source, used as a moment if the time series is thin). The exogeneity claim is that a single facility is a price-taker on fuel margins, so $\tilde{R}$ is plausibly outside its own tank decisions; the standard caveat is that a local demand shock could move both revenue and exit, which is why the cross-sectional county evidence is treated as supporting rather than decisive.

One practical point, also from Sweeting. Retail gasoline prices carry a large national trend, and a stationary model cannot contain calendar time. Sweeting faces the same issue — radio listening and revenue-per-listener both trend over his sample — and handles it by noting that the real (inflation-adjusted) series is roughly flat and having firms treat the current level as the forecast going forward. We do the same: the firm forecasts the persistent, real (detrended) margin through Γ_R, not the raw national price path.

The assumption evidence suite

This section supersedes the open checks in the sketch above. Every empirical claim comes from the rebuilt evidence suite (scripts `AE01`–`AE08`, Ticket 021), purpose-built so that each model assumption has exactly one exhibit, computed on one common sample with one common action definition. Earlier diagnostic versions (the M- and E-series) remain on disk as the audit trail. Each exhibit below carries the model assumption as a centered equation, a brief statement of why the model needs it and how the evidence supports it, then three short paragraphs: why we run the analysis; how exactly it is computed; and what the result says — first in plain terms, then for the model assumption it supports.

Sample and conventions used by every analysis below

All analyses run on the same facility-year panel: every facility-year from 1999 onward with at least one tank at the beginning of the year (2.35 million facility-years; 21 states; Texas is the only risk-priced state). A facility's beginning-of-year (BOY) portfolio is described by counts of tanks over 16 categories: wall type (single- or double-walled) crossed with eight five-year age bins — the same age bins the estimation pipeline already uses. Each facility-year is assigned exactly one of five behaviors, in this order of precedence: **Exit** (every tank closed), **Replace** (closed at least one tank and installed at least one, still operating), **Downsize** (closed at least one tank, installed none, still operating), **Expansion** (installed without closing anything), and **Maintain** (no change). Expansion is counted but never pooled with the model's actions. Analyses that condition on the insurance regime restrict to the estimation sample (Texas 2006+ plus control states); purely physical analyses use the full panel. Every regression reports standard errors clustered by state (18 clusters).

What facilities do in a year

$$\mathcal{A}(s) = \{(k, m)\} \cup \{X\}$$

The model prices four actions: maintain, downsize, replace, exit. This exhibit confirms all four are real, countable behaviors in the data. One note on the model's multinomial structure: the choice among all options is modeled as a single multinomial choice; the evidence regressions below use binary logits (one action versus all others) because clustered standard errors are native there and, with 97.5% of facility-years at maintain, the two approaches are numerically near-identical at the rare margins.

Why. The model cannot price an action that is absent from the data. The table establishes that rare actions are rare the way heart attacks are rare — low annual probability, large absolute counts. It also records expansion separately, justifying its exclusion from the action set.

How it is computed. Each facility-year in the full panel is assigned one behavior using the closure and install flags from the facility panel, with the precedence order above. Counts of tanks removed and installed within the year come from the tank-level BOY build (a tank is "removed in year t" if it was present at the start of t and has a closure event in t; "installed in t" if its install year is t). The table reports event counts and shares of facility-years; the simultaneity row counts facility-years with two-plus removals and two-plus installs.

Table 2: Facility-year behavior frequencies

Action	N	Rate pct
Panel A. Action frequency, all facility-years		
Exit	40,304	1.71
Replace	6,422	0.27
Downsize	7,325	0.31
Expansion	4,409	0.19
Maintain	2,293,497	97.51
Panel B. Simultaneous shed and install		
2+ sheds and 2+ installs	4,295	0.18

Notes: Full 1999+ panel, 2.35M facility-years. Panel A assigns each facility-year one behavior by the precedence order Exit, Replace, Downsize, Expansion, Maintain. Panel B counts facility-years with at least two tank removals and at least two installs in the same calendar year. Per-size columns are in `AE_X1_Action_Frequencies.csv`.

What it says. Facilities do nothing 97.5% of the time — and when they act, full exit (40,304 events) dominates, but downsizing (7,325) and replacement (6,422) are each thousands of events, with downsizing slightly the more common. For the model: every action the model prices has thousands of observations behind its cost parameter, and the 4,295 facility-years that remove and install multiple tanks at once mean even a fixed mobilization cost is not ruled out by the data. The 4,409 expansion years justify fencing expansion off rather than forcing it into a model action.

Size changes what facilities do, holding everything else fixed

$$n \in s$$

Tank counts are a state variable. The exhibit compares behavior within the existing 32 (age, wall, regime) cells: if size shifts behavior within those cells, it must be in the state. The three size odds ratios — 0.48 exit, 27.1 downsize, 3.6 replace — each come from a regression that absorbs every cell fixed effect, so they reflect comparisons of same-cell owners of different sizes.

Why. The current model's state has no notion of facility size: a one-tank and a five-tank facility in the same (age, wall, regime) cell are forced to behave identically. If size shifts behavior within those cells, size must be in the state. This is the single most important fact motivating the portfolio model, so it gets both a regression (the formal statement) and a figure (the raw pattern).

How it is computed. On the estimation sample, for each of the three non-maintain margins separately, we run a binary logit of "did this facility-year take that action" on facility-size indicators (2, 3, 4+ tanks, with 1 tank as the base), including a fixed effect for every one of the 32 (age $\times$ wall $\times$ regime) state cells — so size is compared only within cells, never across them. The table reports odds ratios: an entry of 2 means that action is twice as likely at that size as at a one-tank facility in the same cell. The figure shows the same pattern without any model: raw action shares by age bin, drawn separately by size and wall type.

Table 3: Action odds by facility size, within state cells

Size vs one	Exit	Downsize	Replace
size 2	0.72***	8.12***	1.82***
size 3	0.59***	9.34***	3.30***
size 4 plus	0.48***	27.08***	3.60***

Notes: Binary logit per margin (vs. all other behaviors), fixed effect for each of the 32 state cells, SE clustered by state (18). Odds ratios relative to one-tank facilities; stars: *** p<.01, ** p<.05, * p<.1. Estimation sample; per-margin sample sizes in `AE_X2_Size_Logit.csv`.

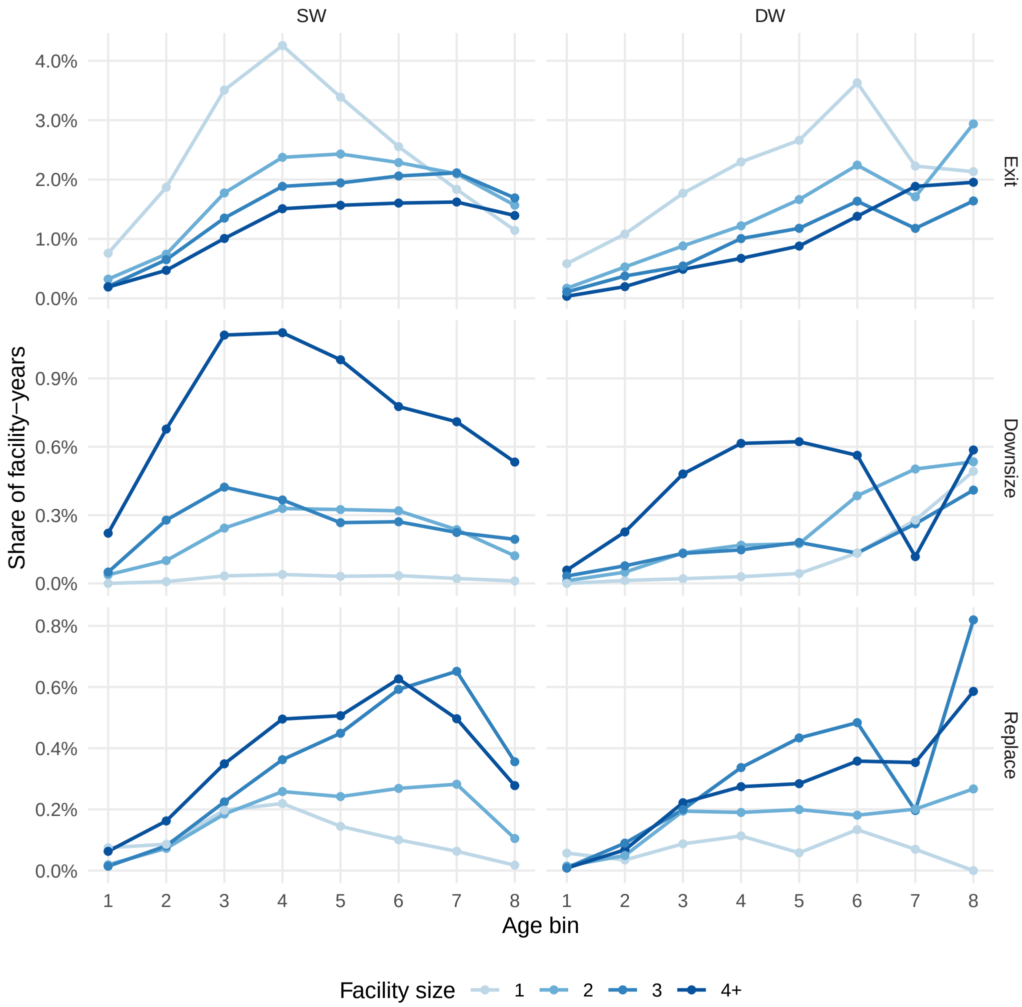

Figure 1: Action shares by age, size, and wall

Notes: Each line is a facility-size class (darker = larger); rows are the three non-maintain behaviors; columns are wall type. Vertical axes differ by row because base rates differ by an order of magnitude. Estimation sample, raw shares with no controls.

What it says. Hold the state cell fixed and move only size: a 4+-tank facility is half as likely to exit as a one-tank facility (odds ratio 0.48) but twenty-seven times as likely to downsize (27.1) and three-and-a-half times as likely to replace (3.6). In plain terms: big facilities don't leave the industry, they prune and recycle their tanks; small facilities can only stay or go. For the model: behavior is not just shifted by size, it is reversed across margins — no reweighting of a sizeless model can reproduce that, so tank counts must be a state variable. This is the load-bearing exhibit for the portfolio state.

A facility's tanks are nearly identical to each other

tanks within a (wall, age band) category are interchangeable

This is bookkeeping, not a claim about behavior: the model's agent cannot tell two same-category tanks apart — and neither can the real owner's bill or risk, since price and hazard are constant within a category. The exhibit measures how often real portfolios violate category boundaries and finds the answer, for most facilities, is almost never.

Why. The portfolio state describes a facility by counts of tanks per (wall, age-bin) category, not tank-by-tank. That description loses information only when a facility's tanks differ from each other within a category boundary — so this exhibit measures how alike a facility's tanks actually are. It is the feasibility check for using 16 categories instead of a full tank-level record.

How it is computed. For each facility-year on the estimation sample we take the oldest and youngest tank ages and the all-single-walled / all-double-walled flags from the facility panel. A facility-year is "age-homogeneous" if its oldest and youngest tank fall in the same five-year bin, "wall-homogeneous" if all tanks share one wall type, and "both" if both hold. The figure plots the share of facility-years passing each test, by facility size.

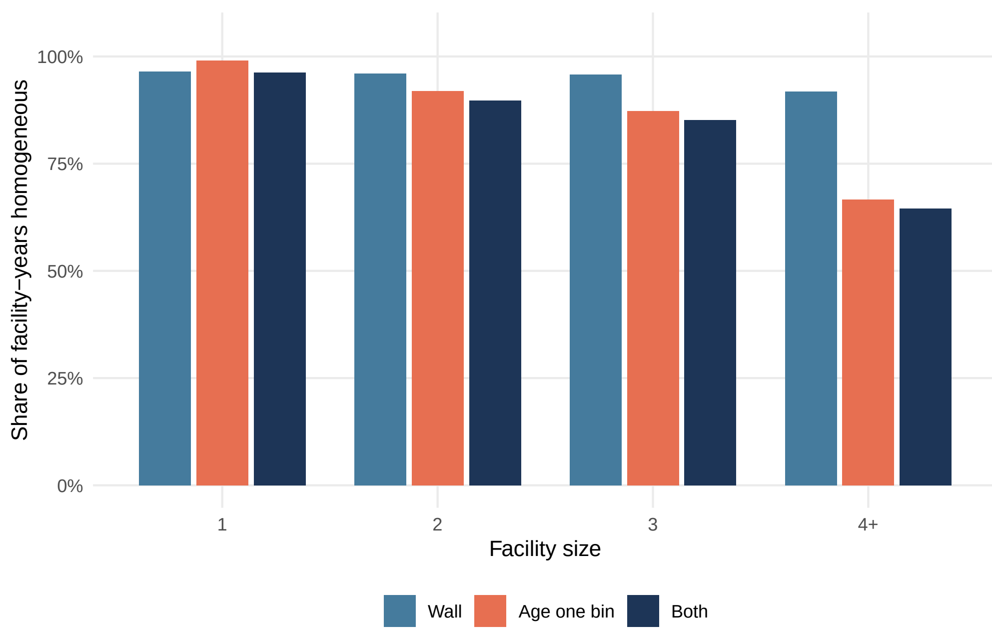

Figure 2: Within-facility tank homogeneity by size

Notes: Bars are the share of facility-years whose tanks are all one wall type, all within one five-year age bin, or both. Estimation sample; counts by size in `AE_X5_Homogeneity.csv`. The median within-facility age range is zero years at every size.

What it says. At the median facility, the age gap between the oldest and youngest tank is zero years — tanks are installed together, in batches. Even at 4+-tank facilities, 64.5% of facility-years have every tank in a single (wall, age-bin) category; at small facilities it is above 90%. In plain terms: a facility is usually "three single-walled tanks from 1988," not a museum of vintages. For the model: describing a facility by category counts is exactly right for the large majority of facility-years and only coarsens the heterogeneous tail of big facilities — which is precisely where counts (rather than one representative tank) still carry the relevant information.

Capacity moves like Rust's mileage

$$G' \sim \Gamma_G(\cdot \mid G,\ m-k), \quad \text{degenerate at } (0,0)$$

Total gallons is the scale variable for operating revenue. The model discretizes it into quartile bins — exactly as Rust discretizes bus mileage — and requires the empirical transition matrix. The two tables below provide that object: the age-bin kernel and the capacity-bin kernel together. The key finding is that the maintain bin is nearly absorbing (diagonal 0.998), while downsize and replace see genuine bin movement (stay 0.564 and 0.465 respectively).

Why. The transition matrix is the law of motion itself. An earlier read took the near-zero median capacity change at replacements as evidence the capacity bin was frozen there too; the transition matrix, which is what the model actually consumes, shows genuine bin-crossing at replace and downsize that the median hid.

How it is computed. Total facility capacity is split at sample quartiles into four bins. The age-transition table records, for consecutive facility-year pairs by action, the fraction of individual tank-events staying in each age band versus advancing. The capacity-transition table records (this year's bin, next year's bin) probabilities separately for Maintain, Downsize, and Replace facility-years.

Table 4: Age-bin transition kernel by action

Age band	Stay	Advance
1	0.725	0.275
2	0.770	0.230
3	0.775	0.225
4	0.795	0.205
5	0.811	0.189
6	0.813	0.187
7	0.828	0.172
8	1.000	0.000

Notes: Rows are beginning-of-year age bands (1–8); columns are the following year's band. Stay probabilities range 0.725–0.828 across bands 1–7; band 8 is absorbing by construction. Computed on consecutive-year tank records; estimation sample.

Table 5: Capacity-bin transition matrix by action

From bin	Q1	Q2	Q3	Q4
Panel A. Maintain				
Q1	0.999	0.001	0.000	0.000
Q2	0.000	0.997	0.001	0.002
Q3	0.000	0.000	0.995	0.005
Q4	0.000	0.000	0.000	1.000
Panel B. Downsize				
Q1	0.995	0.003	0.001	0.001
Q2	0.418	0.574	0.004	0.004
Q3	0.145	0.472	0.372	0.010
Q4	0.069	0.112	0.299	0.520
Panel C. Replace				
Q1	0.997	0.003	0.000	0.000
Q2	0.453	0.547	0.000	0.000
Q3	0.083	0.738	0.173	0.005
Q4	0.015	0.204	0.322	0.459

Notes: Rows are this year's capacity quartile bin; columns are next year's; entries are row probabilities. Estimated separately on facility-years coded Maintain, Downsize, and Replace (consecutive-year pairs, estimation sample). Counts in `AE_X4_G_Transition.csv`.

What it says. Under maintain, the capacity bin is frozen: 0.998 of maintain years stay in their bin. Under the partial actions the bin genuinely moves: only 0.564 of downsize years and 0.465 of replace years stay put. In plain terms: capacity is a fixed attribute until the facility touches its tanks. For the model: the capacity kernel Γ_G enters the law of motion only at downsize and replace; at maintain it is degenerate. The kernel will be re-estimated conditioned on net tank change $m - k$ in the estimator's first stage — a remove-3/install-2 event is a physically different event from remove-4/install-1.

When one tank among several is removed, it is the single-walled one

the k removed tanks are the oldest single-walled; if fewer than k exist, the oldest double-walled fill the remaind

Owners remove the riskiest tank from their portfolio. The conditional logit below compares, within the same facility in the same year, the closed tank against its surviving neighbors. The wall and age coefficients translate directly into the removal rule the model uses.

Why. The model's transition rule for partial actions needs to know which tank leaves. This exhibit estimates, within facilities that removed exactly one of their several tanks, what distinguishes the removed tank from its surviving neighbors — the direct evidence behind the removal rule.

How it is computed. From the tank-level panel we keep facility-years with two or more active tanks of which exactly one closed (19,744 events; 76,626 candidate tanks). A conditional logit compares the closed tank to the surviving tanks of the same facility in the same year — everything facility-level cancels because the comparison never crosses facilities. Regressors: tank age, a single-walled indicator, capacity in thousands of gallons; SE clustered by state.

Table 6: Which tank is removed: within-facility comparison

Closed tank attribute	OR	SE	p
age per year	1.044	0.009	0.000
SW	2.853	0.314	0.001
capacity per 1000 gal	1.000	0.000	0.847

Notes: Conditional logit; each removal is compared only against the same facility-year's surviving tanks. 19,744 events, 76,626 tank-alternatives, SE clustered by state (18).

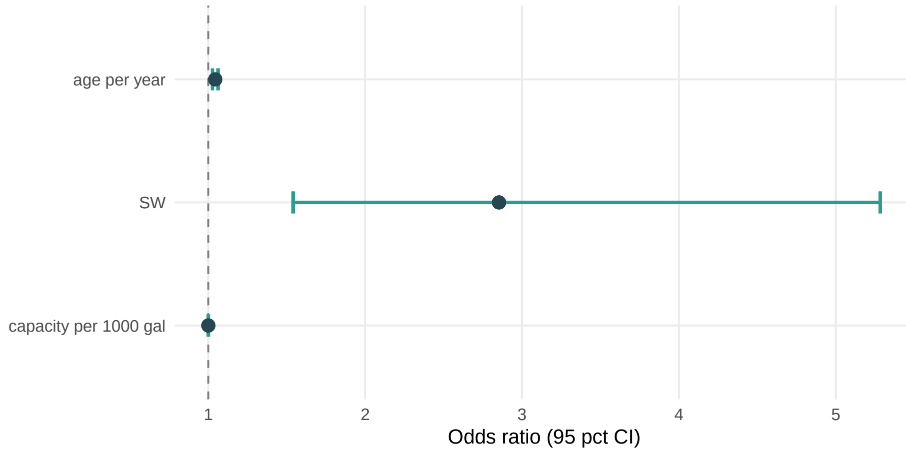

Figure 3: Removal odds ratios, within facility

Notes: Points are odds ratios from the conditional logit in the table above; bars are 95 percent confidence intervals on state-clustered standard errors. The dashed line at one marks no effect.

What it says. Within the same facility, the removed tank is single-walled at 2.85-to-1 odds, each year of age adds 4.4% to its removal odds, and its size is irrelevant (odds ratio 1.000). In plain terms: when an owner culls one tank from the herd, they cull the risky construction type, with a mild preference for the oldest — never simply the smallest. The wall coefficient is worth about 24 years of age, which is why the rule puts every single-walled bin ahead of the double-walled ones. For the model: this justifies a removal rule stated on exactly the two dimensions the state tracks (wall, then age) and justifies leaving capacity out of the rule.

The rule predicts which tanks leave

$r(n, k)$ is the identity rule; k and m are choices

The rule correctly names the removed categories in 84% of events; its load-bearing half — single-walled before double-walled — fails in only 2.8%. The miss-cost table closes the remaining worry quantitatively: even when the rule misses, the pricing and hazard consequences are small. This closes the big-facility downsizing worry quantitatively.

Why. The model assumes removals follow the stated rule. This exhibit checks the rule against every real removal event in the units the model uses — which (wall, age-bin) categories lost tanks — and separately asks how many tanks facilities remove at once. The two questions are distinct: the rule fixes which tanks leave given their number; the number is a choice (see X7b).

How it is computed. For every Downsize or Replace facility-year we know how many tanks were removed (k) and from which categories. The rule's prediction is mechanical: walk the categories in priority order (single-walled oldest-to-youngest, then double-walled likewise) and take the first k tanks the facility owns. An event is an exact match if predicted and actual categories coincide. Two failure types are tallied: removing any double-walled tank while keeping a single-walled one (a wall miss — the part the economics leans on), and removing from a younger bin while keeping an older same-wall bin (an age miss). The miss-cost table computes, for events where the rule mispredicts, the implied premium error and hazard error.

Table 7: Removal-rule fidelity by action and size

Action	Size	N	Match	Wrong wall
Downsize	1	62	1.000	0.000
Downsize	2	1,060	0.765	0.042
Downsize	3	1,486	0.775	0.036
Downsize	4+	4,405	0.697	0.054
Replace	1	467	1.000	0.000
Replace	2	1,034	0.985	0.003
Replace	3	2,454	0.987	0.003
Replace	4+	2,392	0.929	0.013

Notes: Match is the share of events whose removed categories exactly equal the rule's prediction. Wrong wall is the share removing a double-walled tank while keeping a single-walled one; remaining misses are age-bin misses, almost all one bin away. Per-k detail in `AE_X7_Removal_Fidelity.csv`; removed-count vs. top-priority-group comparison in `AE_X7_Shed_vs_Block.csv`.

Table 8: Premium and hazard error at rule misses

Action	Size	Misses	Mean premium err	p90 premium err	Mean hazard err pp
Downsize	2	249	58	124	0.263
Downsize	3	334	52	106	0.144
Downsize	4+	1,333	77	159	0.109
Replace	2	15	60	145	0.312
Replace	3	31	55	106	0.151
Replace	4+	171	85	195	0.123

Notes: Among events where the rule mispredicts the removed category, premium error is the difference between the predicted and actual per-tank premium implied by the composition; hazard error is the analogous difference in facility hazard probability. Card-based premium errors are upper bounds; the empirical card is flatter.

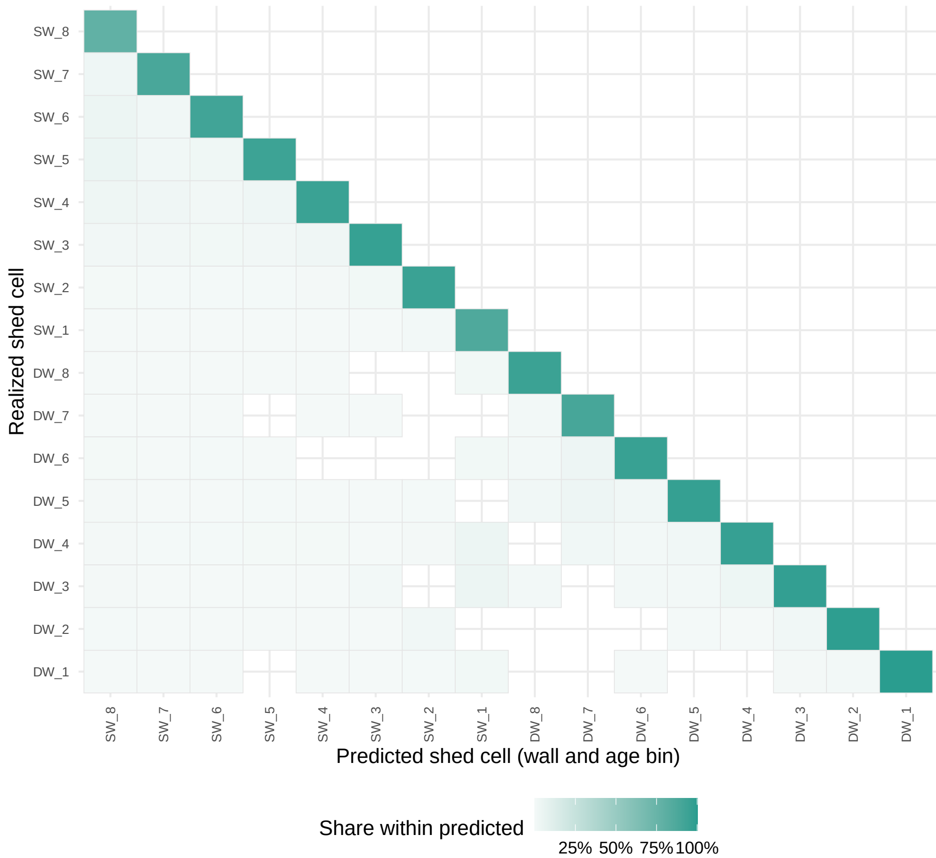

Figure 4: Predicted versus realized removal category

Notes: Rows and columns are the 16 (wall, age-bin) categories ordered by removal priority; shading is the share of removals the rule predicted in the column category that actually came from the row category. 90.5 percent of all removed tanks sit on the diagonal.

What it says. The rule names the exact removed categories in 84% of all events, and its load-bearing half — single-walled before double-walled — fails in only 2.8%. The heat map shows the misses sit one age bin off the diagonal. The miss-cost table closes the big-facility downsizing worry quantitatively: conditional on a miss, the mean premium error is \$71/yr (p90 \$142) on annual bills of \$500–\$1{,}065, and the hazard error is 0.14 percentage points; unconditionally the miss cost is approximately \$11/yr — 1–2% of the bill. These are upper bounds because the empirical premium card is flatter than the filed card. The wall part of the rule holds equally well for replace and downsize (3–6% misses across both); even the 16% of events where the age part misses impose negligible pricing consequences. For the model: the removal identity is deterministic and its errors are small enough to treat as measurement noise, not structural misspecification.

How many tanks: a chosen intensity

$$k,\ m \text{ are choices; only } r(n,k) \text{ is a rule}$$

The count of tanks removed (k) and installed (m) is not predicted by any rule — it is the chosen intensity of the action. The menu works like wings priced per wing: each (k, m) pair is just another option, and two per-unit costs price every line; the model never predicts k or m, the owner chooses them.

Why. The number of tanks involved in a replacement or downsize is the object priced by c^{rem} and c^{inst}. Confirming it varies freely over the data, rather than concentrating at one-for-one swaps, justifies per-unit costs rather than a fixed-menu price.

How it is computed. For every Replace and Downsize facility-year, the table cross-tabulates the count of tanks removed (k) against the count installed (m, zero for downsize events).

Table 9: (k, m) cross-tabulation: tanks removed and installed per event

Tanks removed	Install 0	Install 1	Install 2	Install 3	Install 4 plus
0	360	19	6	2	0
1	4,642	668	120	43	24
2	1,173	355	590	132	57
3	758	431	1,164	577	194
4 plus	612	239	658	512	411

Notes: Replace events (rows $m \geq 1$) and Downsize events ($m = 0$). Counts of facility-years. Only 13.7% of replacements are one-for-one swaps; the modal replacement removes 3 tanks and installs 2. Detail in `AE_X7_KM_Menu.csv`.

What it says. Only 13.7% of replacements remove exactly one tank; the modal replacement removes 3 and installs 2. Downsize events span a similarly wide range of k values. In plain terms: replacement is not a one-tank event and its scale is not fixed. For the model: two per-unit cost parameters (c^{rem} and c^{inst}) price a wide menu of intensities, and their identification comes from the variation in (k, m) counts visible here.

Replacement recycles capacity; it does not reset the facility

$$n' = n - r(n,k) + m\, e_{DW,1} \quad \text{(no reset)}$$

This exhibit exists to kill the old reset-to-zero replace transition and pin "tanks out, tanks in, scale carried." The current 6p model's replace transition rebuilds the facility from scratch — and 98% of real replacements land somewhere that transition says is impossible. The new transition carries the portfolio along.

Why. If replacement reset the state, the post-action composition would be deterministic and there would be no role for capacity or age heterogeneity surviving a replacement. The data show replacement is recycling — the facility keeps its scale and refreshes a tank.

How it is computed. The population is facility-years that closed at least one tank and kept operating — nothing else. Events split into "swap" (at least one install that year) and "shrink" (none). For each event we compute the percent change in total capacity, the net change in tank count, and — at the tank level — whether a single-walled tank was removed and a double-walled one installed.

Table 10: What replacement does to the portfolio

Event	Count chg	N	Share	Med cap chg	Shed SW inst DW
swap	-1	2,037	0.352	0.0	0.620
swap	-2	1,137	0.196	-14.3	0.637
swap	0	2,133	0.368	10.0	0.527
swap	-3 or more	482	0.083	-25.0	0.591
shrink	-1	4,786	0.653	-10.7	0.000
shrink	-2	1,202	0.164	-42.0	0.000
shrink	0	0	0.000	NA	NA
shrink	-3 or more	1,337	0.183	-66.7	0.000

Notes: Closure-years only (operating, at least one tank closed). Swap = at least one install the same year. Count chg is the net tank-count change; Med cap chg the median percent capacity change; Shed SW inst DW the tank-level share removing single-walled and installing double-walled. Swap-minus-shrink differences: capacity +84.4 points (SE 15.7), count +0.91 tanks (SE 0.15), state-clustered (`AE_X8_Recycle_Tests.csv`).

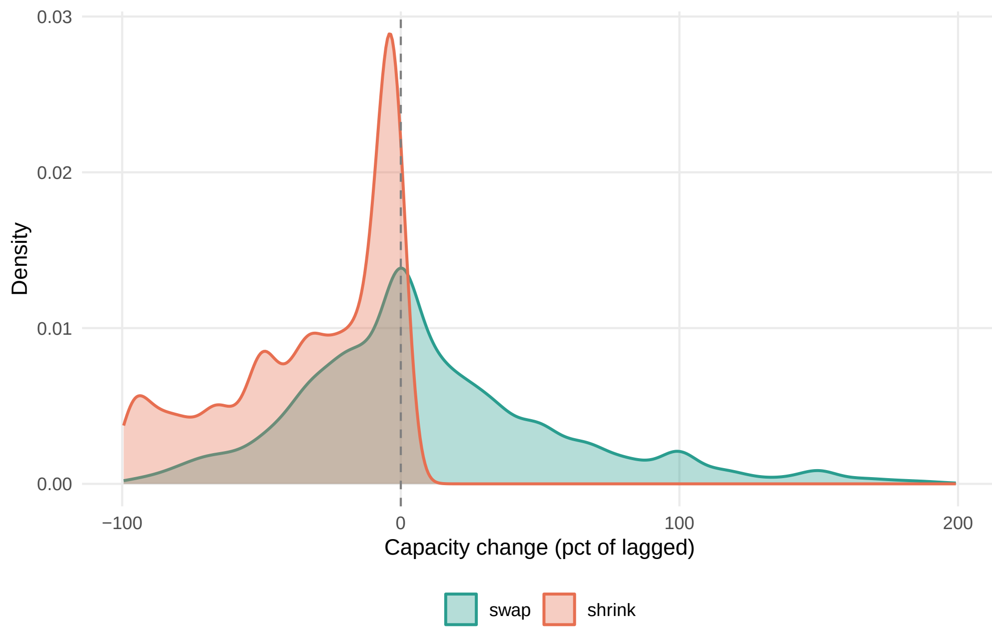

Figure 5: Capacity change at swap versus shrink events

Notes: Densities of the percent change in total facility capacity over the event year, for swap (removed and installed) and shrink (removed only) events. Display trimmed to $(-100, 200)$ percent; closure-years only.

What it says. When a facility swaps, its capacity barely moves — the median change is $+6.9\%$, 0.0%, and -9.7% at net count changes of $0, -1, -2$ — while shrink-only events destroy capacity (-14%, -39%, -58% at the same count changes). Swapping adds 84 percentage points of capacity retention relative to shrinking. In plain terms: replacement is recycling — the facility keeps its scale and refreshes a tank. For the model: the replace transition removes one tank by the rule and adds one new tank, carrying the portfolio along; nothing resets.

New tanks are double-walled — by regulation

$$\text{installs} = e_{DW,1}$$

Every new tank placed in the model enters as a double-walled unit in age bin 1. This is an institutional fact: every sample state has banned new single-walled installs, and the installs that earlier looked like rule violations were almost entirely legal pre-ban events.

Why. The model's replace action installs a double-walled tank, with no choice about it. This exhibit shows that assumption is regulation, not preference, and pins the 6% post-ban single-walled install rate as a registry wall-coding issue rather than genuine violations.

How it is computed. Among Replace facility-years at facilities holding at least one single-walled tank, each event is classified at the tank level into three exhaustive classes: removed single-walled and installed double-walled (the rule); removed single-walled and installed single-walled (legal only pre-ban); removed double-walled only (a genuine violation). Each event is stamped pre- or post- its own state's verified ban date. The figure plots the double-walled share of all installs by year, Texas versus control states, with tick marks at the ban dates.

Table 11: Replacement install decomposition by ban timing

Install class	Ban timing	N	Share pct
shed SW inst DW	post ban	2,402	41.40
shed SW inst DW	pre ban	1,365	23.53
shed SW inst SW	post ban	156	2.69
shed SW inst SW	pre ban	1,804	31.09
shed DW only	post ban	37	0.64
shed DW only	pre ban	38	0.65

Notes: 5,802 Replace events at facilities holding at least one single-walled tank. Classes are tank-level acts, mutually exclusive and exhaustive. Ban dates are the email-verified state list (Texas 2009-01-01). Shares are of all 5,802 events.

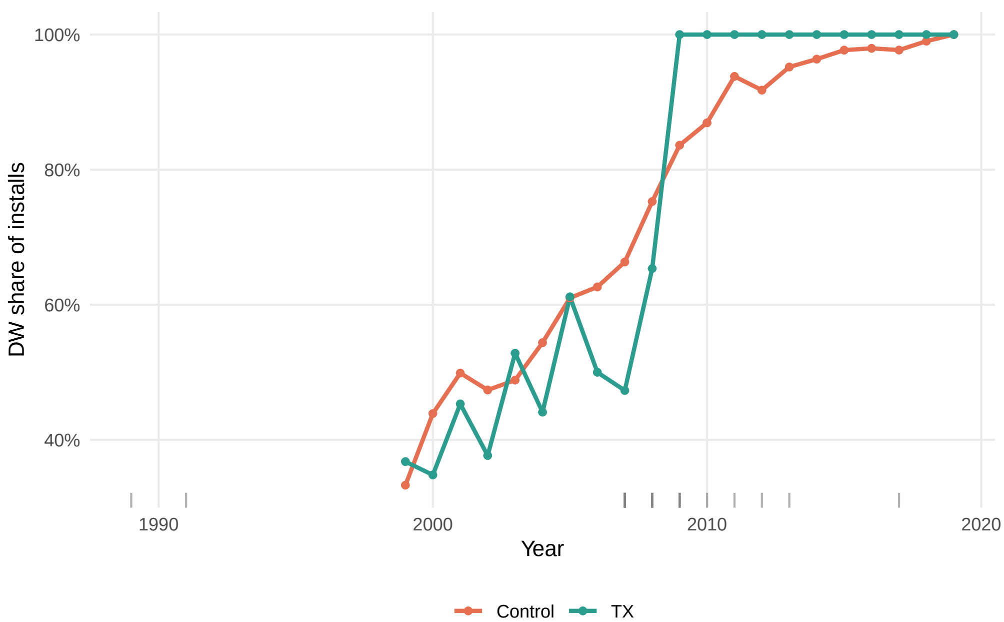

Figure 6: Double-walled share of new installs

Notes: Double-walled share of all tank installs by calendar year, Texas versus control states. Bottom tick marks are state single-walled-install ban effective dates (1989–2017; most 2007–2009).

What it says. Single-walled installs at replacing facilities collapse from 31.1% of events before the installing state's ban to 2.7% after — the regulation bites visibly in the figure as every state's line saturates at 100%. Genuine violations of the removal rule (replacing a double-walled tank while keeping a single-walled one) are 1.3% of events, pre- and post-ban alike. For the model: "the new tank is double-walled" is an institutional fact; the model stays stationary because no counterfactual we run changes the ban. The 6% of post-ban events still coded single-walled-install is flagged as a registry wall-coding issue, not behavior.

The pricing regime moves the action margins

$$P(\text{action} \mid s,\, \rho) \text{ differs by regime}$$

Under risk-based pricing, downsizing falls by half, upgrade-replacement roughly doubles, and exit rises — three margins, three different responses. These are the margins the counterfactuals move (Part 4).

Why. If downsizing and replacing were interchangeable bookkeeping, the insurance regime should not move them differently. This exhibit tests whether risk-based pricing shifts which margin facilities use — the evidence that the margins are economically distinct actions worth pricing separately.

How it is computed. Three separate binary logits on the estimation sample, each with year fixed effects and (age bin, wall, size) controls, SE clustered by state. Downsize and exit use the full sample with a risk-based regime indicator. The replace margin is estimated only on facilities holding a single-walled tank — there, replacing is upgrading by the removal rule, so a single regime coefficient answers the question. (An earlier version interacted regime with wall type and produced unstable coefficients from a nearly empty cell; this design avoids the interaction.) Texas is the only risk-priced state, so these are descriptive contrasts; causal regime claims rest on the difference-in-differences work elsewhere.

Table 12: Regime contrasts by action margin

Margin	Sample	OR	SE	N	States RB
Downsize	full	0.490***	0.184	2,193,723	1
Replace	has SW	2.138***	0.226	1,775,516	1
Exit	full	1.634***	0.162	2,193,723	1

Notes: Binary logit per margin with year fixed effects and age/wall/size controls; SE clustered by state (18). Replace row estimated on facilities holding at least one single-walled tank. RB present in one state (Texas); treat as descriptive. The Texas 2009 install ban partially overlaps the upgrade margin after 2009.

What it says. Under risk pricing, downsizing falls by half (odds ratio 0.49), upgrade-replacement at single-walled facilities roughly doubles (2.14), and exit rises (1.63) — three margins, three different responses, all significant with year effects absorbed. In plain terms: when old risky tanks become expensive to insure, owners stop quietly shrinking and start either upgrading the risky tank or leaving altogether. For the model: an action set that folded downsizing into "maintain" (the current model) hides a margin the policy lever demonstrably moves; the four-action set prices all three responses, which is exactly what the counterfactuals need.

What facilities actually pay

$$P(n, \rho) = \sum_c n_c \, \bar{p}_c \ (RB), \quad \tau_s N(n) \ (FF)$$

The model prices portfolios using the average premium facilities actually paid per cell — not the filed card. Three exhibits establish this object: first, the facility-level picture of how total bills relate to portfolio age; second, the comparison of filed and empirical cards; third, the identification scatter showing premium variation beyond hazard that separates $\hat{\gamma}_p$ from $\hat{\gamma}_r$.

Why. The model prices a portfolio by summing a per-tank price over its (wall, age-bin) categories, so the pricing exhibit should be exactly that object measured in the data. The filed rate card is shown beside it because the comparison is itself a finding: the card is steep in age, but realized bills are much flatter.

How it is computed. *Figure 1 (facility totals by age):* For Texas single-category facility-years (all tanks in one (wall, age-bin) cell, which are 86.6% of Texas facility-years), total annual premium is plotted against the common age bin. Flat-fee states' total bills are plotted at their state's per-tank fee times tank count. Note: older portfolios hold approximately twice as many tanks on average; part of the total-bill gradient is count, not per-tank price. Mixed-composition facilities have no single age bin and so are excluded from this figure — which is precisely why the model prices by composition rather than average age. *Filed and empirical card tables:* The filed table evaluates the 2006 filed schedule for a standard tank in each cell; the empirical table averages the engine-rated per-tank premium among single-cell Texas facility-years (245,578 observations) within each cell. *Scatter:* for every distinct observed composition, the average premium index against facility hazard $H(n)$ (ML model at portfolio composition), one point per composition sized by facility-years.

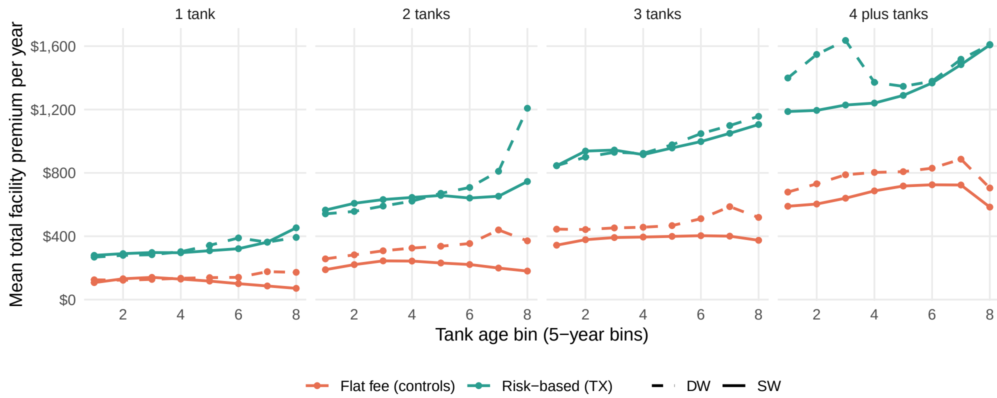

Figure 7: Total facility premium by age band

Notes: Texas single-category facility-years (all tanks in one (wall, age-bin) cell; 86.6% of Texas facility-years); flat-fee control-state totals plotted at state fee × tank count. Single-walled TX totals rise from approximately $500 to $1,065 across age bands; flat-fee lines are flat by construction. Older portfolios hold ≈2× more tanks on average — part of the gradient is count, not per-tank price. Mixed-composition facilities are excluded because they have no single age bin, which is why the model prices by composition.

Table 13: Filed rate card, dollars per tank-year (2006 era)

Wall	1	2	3	4	5	6	7	8
SW	319	354	372	389	425	442	442	442
DW	248	283	301	319	354	372	372	372
2014 and 2019 re-filings raise old-age loads; era table in qmd notes								

Notes: Columns are the eight five-year age bins. Standard tank: $300 base, statutory liability factor 1.18, age load by bin, double-walled credit −0.20; equipment, status, and deductible factors at neutral defaults. Later re-filings extend old-age loads upward (`AE_X3_RateCard.csv`).

Table 14: Average premium actually paid, by cell (2006 era)

Wall	1	2	3	4	5	6	7	8
SW	281	295	302	289	297	310	300	278
(facility-years)	2,955	8,128	13,101	14,494	22,059	20,941	10,707	8,755
DW	265	267	277	267	280	306	320	288
(facility-years)	5,526	7,851	8,436	5,546	3,079	1,023	381	206

Notes: Average engine-rated per-tank premium among single-cell Texas facility-years (all tanks in one (wall, age-bin) category), 2006-era years; facility-year counts beneath each average. Old double-walled cells (bins 7–8) are thin (n=381/206) — the DW crossover in those bands is noise and selection, not a genuine age-load reversal. All eras and dispersion in `AE_X3_Empirical_Premium.csv`.

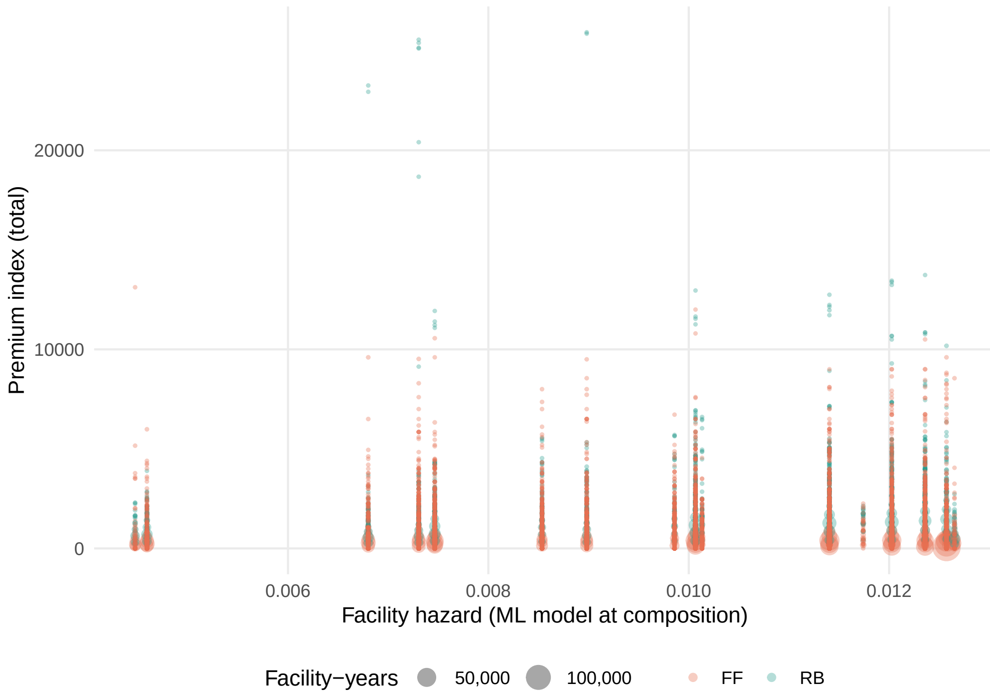

Figure 8: Two pricing worlds over one state

Notes: Premium index against facility hazard $H(n)$ (ML model at portfolio composition), one point per observed portfolio composition; point size is the number of facility-years with that composition. Teal: risk-priced (Texas). Orange: flat-fee control states, where the bill is the per-tank fee times the count and is zero in the six states charging operators nothing.

What it says. The filed card climbs from \$319 to \$443 per tank-year as a single-walled tank ages, with a \$71 double-walled discount. What facilities actually pay climbs barely at all: single-walled cells average \$278–310 across every age bin, double-walled \$265–320, and the realized wall gap is \$15–30. In old double-walled bins 7–8 the crossover is noise and selection — those cells are thin (n=381/206 facility-years). A candidate explanation: out-of-service status credits on old tanks and equipment credits offset the age loads in practice. For pricing: $\bar{p}_c$ will be the empirical averages, not the filed card rates. For identification: $\hat{\gamma}_p$ leans less on the compressed cross-cell gradient and more on the regime gap (\$708.8 at matched composition; X11) and the re-filings that moved the schedule over time.

Is there premium variation beyond hazard and size?

$$\mathrm{Corr}(P,\, H \mid \text{controls}) < 1$$

Within risk-priced Texas, 26.7% of the premium index's variation survives controls for hazard, tank count, capacity, and year; at matched composition, risk pricing costs a facility \$708.8

more per year (SE \$127.5). These are the corrected-H estimates, where H is the facility-level hazard from the ML model evaluated at the actual portfolio composition. The residual variation is what separates $\hat{\gamma}_p$ from $\hat{\gamma}_r$.

Why. Price sensitivity and risk sensitivity are separate parameters, and they can be told apart only if premiums move independently of hazard. This table asks the question directly on the constructed premium index.

How it is computed. Each facility-year's premium index is built from its actual composition: under risk pricing, the per-cell price summed over the facility's tanks; under flat fees, the state's per-tank fee times the tank count. Regressions of the index on portfolio hazard $H(n)$, tank count, capacity, and year fixed effects run separately for the risk-priced sample, all flat-fee rows, fee-positive flat-fee rows, and pooled (the pooled row adds the regime indicator: the premium gap at matched composition). Zero-fee flat-fee states: Kansas, Maryland, Minnesota, Oklahoma, South Dakota, and Virginia (40.2% of flat-fee facility-years).

Table 15: Premium regressions: residual variation and the regime gap

Spec	R2	Resid SD	Resid over SD P	Coef regime	SE regime	N
RB	0.929	148.0	0.267	NA	NA	426,791
FF all	0.118	511.1	0.939	NA	NA	1,925,166
FF feepos	0.358	476.7	0.801	NA	NA	1,151,584
pooled	0.333	497.2	0.817	708.8	127.5	2,351,957

Notes: Year fixed effects throughout; SE clustered by state except the RB row (one state). Resid/SD(P) is the residual SD as a share of the premium SD — the variation left after hazard, count, capacity, and year. Hazard is the facility-level ML-model construction at portfolio composition. Zero-fee flat-fee states: KS, MD, MN, OK, SD, VA (40.2% of flat-fee facility-years). Per-tank engine fit by era in `AE_X11_RateCard_Fit_PerTank.csv`; correlations in `AE_X11_Correlations.csv`.

What it says. Within risk-priced Texas, 26.7% of the premium index's variation survives the controls — the card's discrete age steps and re-filings move the bill in ways hazard and size cannot mimic — and at matched composition, risk pricing costs a facility \$708.8 more per year (SE \$127.5). In plain terms: there are two genuinely different prices for the same physical facility, and even within the risk-priced world the schedule has kinks that risk itself does not track. For the model: that is the variation that separates $\hat{\gamma}_p$ from $\hat{\gamma}_r$.

The state space is small in practice

368 compositions cover 95% of facility-years

The portfolio state is defined over a large combinatorial space, but real facilities are small and their tanks are alike. The coverage curve below shows 368 portfolio types cover 95% of facility-years, and 86.6% of facility-years are a single-category facility. Computation and data coverage are non-issues.

Why. The portfolio state sounds enormous (every possible count vector over 16 categories). This exhibit shows it is not: a few hundred portfolio types describe nearly everything.

How it is computed. Every facility-year's composition is written as its 16-category count vector; identical vectors are pooled and ranked by how many facility-years share them. The curve plots the cumulative share of facility-years covered against the number of distinct compositions, log scale.

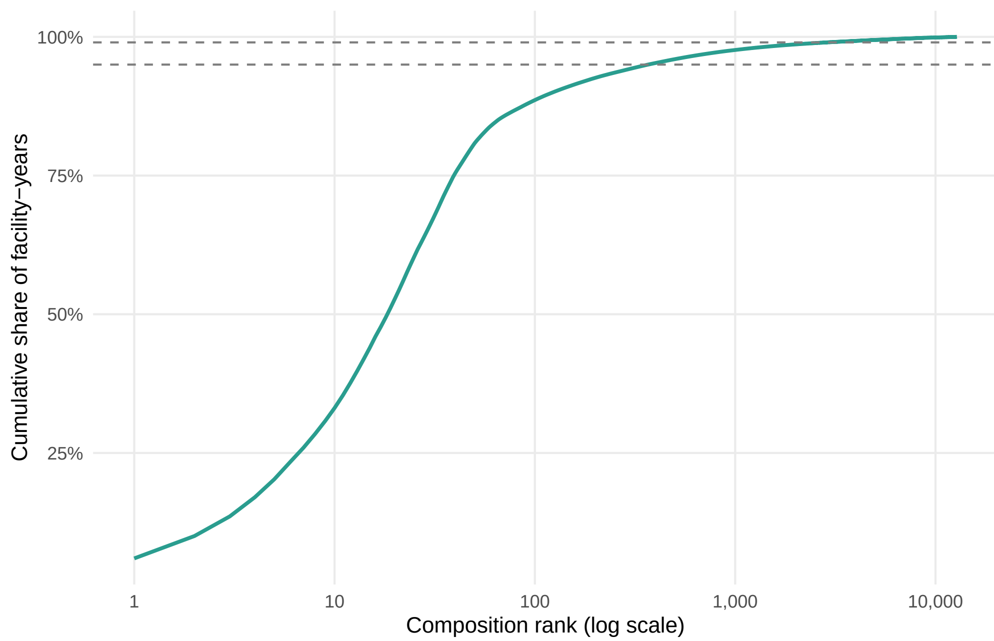

Figure 9: Coverage of facility-years by composition rank

Notes: Cumulative share of facility-years covered by the most common portfolio compositions, by rank (log scale). Dashed lines at 95 and 99 percent. 12,781 distinct compositions observed; 368 cover 95 percent of facility-years; 86.6 percent of facility-years occupy a single category.

What it says. Just 368 of the 12,781 observed portfolio types cover 95% of all facility-years, and most facility-years (86.6%) are a single category — "n tanks, all alike." In plain terms: the state space is big on paper and small in life. For the model: the value function is cheap to compute, every commonly visited state has abundant data behind its choice probabilities, and the exotic compositions the model can represent but the data rarely visits matter only in counterfactuals — which is exactly when you want the model, not the data, to extrapolate.

What the counterfactuals need, and how far the data lets them go

Each counterfactual is a surgical change to one object in the model. For each, the paragraphs below describe the surgery, the margins it activates and which exhibits cover them, and the data resolution the result carries.

CF1: Texas stays flat-fee

The surgery. Transplant the whole contract $C = (\tau, D)$ — not just the premium level but also the deductible — from the risk-based world to a flat-fee world. The bound is realized through real control-state contracts: the lower bound is a zero-fee state (zero per-tank fee but still carrying its own deductible — CF1-low is not "insurance vanishes"); the upper bound is the highest-fee control contract; the mean contract is the central estimate. The deductible transplant moves the OOP term as well as the premium, so both the γ_p and the γ_r channels are priced in the counterfactual.

Margins and evidence. The CF runs through the regime sensitivity of all three non-maintain margins (X10): how downsize, replace, and exit respond to the premium and deductible levels in the flat-fee contracts. The size-dependent direction of the response (X2) is the key heterogeneity the CF must reproduce. Premium identification (X11) and the pricing object (X3) underlie $\hat{\gamma}_p$; hazard and deductible variation underlie $\hat{\gamma}_r$.

Bounded, not point-identified — Texas's counterfactual flat-fee contract is unobserved, so we report the range spanned by actual control-state contracts.

CF2: First-best contract

The surgery. Set the facility's total private exposure equal to $H(n)(L + E)$ at the margin — the social marginal cost of its hazard — and find the contract that implements this. The welfare wedge derivation in Part 2 specifies the wedge under each regime: the FF wedge is $(L-D)+E$ per unit hazard; the RB wedge is E plus the under-pass-through of $(L-D)$ from the compressed empirical card. The γ_p versus γ_r distinction matters: the instrument that delivers a dollar of Pigouvian correction is not neutral — a premium-channel dollar acts through γ_p and a deductible-channel dollar through γ_r, and if the two are not equal the first-best contract form is not indeterminate.

Margins and evidence. The CF requires that $H(n)$ respond to the contract at the margin, which is covered by X10 and X11. The external damage calibration (E = \$50,000) is the binding assumption — the model does not estimate E from the data.

E is calibrated, not estimated; welfare reported at the calibration and a band.

CF3: Single-walled replacement subsidy

The surgery. Lower c^{inst} (or equivalently subsidize the (k, m) options that include removal of a single-walled tank). The subsidy runs through the full (k, m) menu, activating the replace margin for facilities currently on the maintain or downsize margin. The menu evidence (X7b) shows the relevant intensity range: the modal replacement removes 3 and installs 2, so the subsidy applies to a wide range of events. The removal rule (X6, X7a) identifies which tanks leave and thus which composition improvements the subsidy delivers.

Margins and evidence. The response depends on how many facilities are close to the replacement margin (X1, X7b), how the rule governs which tanks go (X6, X7a), and how the nest parameter λ apportions new replacements between firms that would have acted anyway and firms that would not have. When the subsidy creates new replacements, the model must apportion them between firms that would have removed tanks anyway and firms that would have done nothing. One estimated parameter governs that split, so we report the subsidy's effects both at its estimated value and with it switched off — showing the reader how much of the conclusion rides on it.

The model reports subsidy effects at the estimated $\hat{\lambda}$ and with $\lambda = 1$ (flat logit); the gap between those two is what rides on the nest parameter.

CF4: Age-A removal mandate

The surgery. Delete the options in $\mathcal{A}(s)$ that would leave a tank in age bins above A in the post-action state n'. Because the state tracks tank counts by age bin, the mandate is feasible to implement in the model: the policymaker can observe and enforce the bin count, and the facility can comply either by removing old tanks (any (k, m) with $k \geq n_{old}$) or by the cheap compliance path of removing old tanks without replacement (pure downsize, $(k, 0)$). The cheap compliance margin is present in the model — and in the data (X7b shows downsize events span a wide range of k) — which means the CF will find cost-minimizing compliance paths, not just wholesale replacement.

Margins and evidence. The mandate's risk benefit is measured by how much $H(n)$ falls when old-tank count is reduced (X4 aging kernel). The compliance cost runs through the removal and exit margins (X1, X2, X7).

The mandate's risk benefit is measured at the resolution of the leak data — facility-level — so the model credits removing an old tank with the facility's hazard improvement, not a tank-specific one. No estimate from this data can do better.